Left Behind, Not Averaged Out: A Poset-Based Identification of Multidimensional Youth Deprivation in 24 European Countries

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AI use AI tools were used to support code development. All methodological choices, data checks and the final content were reviewed by, and are the responsibility of, the authors.

Abstract

European countries support young people very differently, and composite indices, which average multiple indicators, can mask this by letting good performance on one dimension compensate for weakness on another. This study applies partial order set theory (poset), a non-compensatory alternative, to identify which of 24 European OECD countries are unambiguously "left behind" in youth deprivation, using five indicators (poverty, NEET, unemployment, low math skills, low upper-secondary completion) from OECD, Eurostat, PISA, and Global Findex data. Each country's identification degree, the share of linear extensions in which it ranks worse than a medium-deprivation threshold, is computed via two independent implementations that agree closely ($\rho = 0.9861$). Five countries are certainly deprived, nine certainly not, and ten remain ambiguous. The poset ranking correlates only moderately with a composite index ($\rho = 0.85$), with countries like France and the UK classified very differently depending on the method. Robustness checks confirm the extreme groups are stable, while intermediate cases are sensitive to indicator choice; NEET is the dimension most often responsible for uncertain classification. A definitive classification is possible only for a subset of countries, and the choice of aggregation method has real policy implications.

Introduction

Over the years, European countries have recognized young people as a political priority and a fundamental resource. However, the intensity and instruments of this attention have varied considerably from one country to another: countries such as Germany and Austria have historically developed a dual vocational education and training system that integrates school and work from adolescence, while others, such as France, have focused more heavily on targeted measures to combat early school leaving and support young NEETs. The result is a fragmented picture, with countries that may appear similar at first glance but, upon closer analysis, reveal different challenges, such as youth unemployment or early school leaving. Comparing the level of support a country provides to its young people is therefore never simple and often depends on the extent of the disadvantage being considered.

To study this question empirically, a data source is therefore needed that is collected using a consistent methodology across countries, designed for international comparison, and based on a globally shared definition. The OECD (Organisation for Economic Co-operation and Development) is an organization composed of multiple countries that produces harmonized statistics specifically to allow national figures to be compared without being affected by country-specific peculiarities. For the study in question, the data come from the Sustainable Development Goals (SDGs), which translate the objectives of the United Nations' 2030 Agenda into concrete indicators. Using data derived from the SDGs and anchored in the UN framework provides a set of indicators that lends greater legitimacy to international comparisons, relying on a globally established standard.

The relevance of this comparison is not only statistical, but also economic and social. Young people represent the future workforce and the demographic foundation upon which the sustainability of European welfare systems depends, in a continent that is already facing a progressive ageing of its population. A generation excluded from the labour market, lacking adequate skills, or leaving the education system prematurely creates not only an individual cost but also a collective cost that persists over time, increasing the risk of intergenerational inequality. Understanding which countries are able to effectively integrate their young people and which leave them behind, and along which specific dimensions, is therefore a question that concerns not only those working on youth policies in the narrow sense, but also the broader economic and social sustainability of Europe as a whole.

The choice to analyze the condition of young people in Europe stems from a methodological rather than thematic observation. Most international comparisons on these issues are based on composite indices, that is, on a single score obtained by averaging multiple indicators. Although this approach may be easier to communicate, it presents a significant conceptual limitation. For example, a country with high youth unemployment but low poverty may obtain the same average score as a country with mediocre performance on both dimensions, despite the fact that these represent very different situations in terms of the policies that should be adopted. This study therefore aims to assess whether, by applying an alternative method, the theory of partially ordered sets (poset), which does not allow for compensations between dimensions, the picture of European countries changes substantially compared with that provided by a traditional composite index. For these reasons, the study is relevant to several stakeholders:

- European and national policymakers are responsible for deciding where to concentrate resources and targeted interventions. Knowing not only which countries are overall the most disadvantaged, but also the specific dimensions in which they are lagging behind, provides information that a composite index tends to conceal, whereas the poset approach makes it explicit.
- Non governmental organizations working on youth inclusion can use the results to more effectively target and prioritize their interventions.
- Researchers and statistical institutions can benefit from the study as it provides a replicable example of applying the poset method to a real world SDG dataset, including cross validation between a from scratch implementation and the reference parsec package. This can be useful for those seeking to assess the reliability of the method when applied to real world data.

Research questions

Based on what has been discussed above, the research questions guiding this study can now be formulated as follows:

How easy is it to figure out exactly which European countries are leaving their young people behind the most, and to what level does this answer depend on the method chosen to combine the five dimensions of deprivation considered: poverty, NEET, unemployment, mathematical skills, and school completion? Three sub-questions follow, each corresponding to a specific step of the analysis.

1. Which countries can be considered unambiguously "left behind," and which cannot, regardless of how the five dimensions are combined? Is there a group of countries whose youth disadvantage falls below a reference threshold under every reasonable way of looking at the data, and, symmetrically, an equally solid group whose condition is certainly better?
2. Which countries are classified substantially differently depending on whether a composite index or a non-compensatory method such as the poset is used, and why? Does the aggregation method significantly alter the judgment on specific countries, or do the two approaches converge, differing only in marginal detail?
3. How well do these results hold up against the methodological choices made throughout the analysis (which indicators to include, where to set the thresholds, how many levels to use), and which single dimension weighs most heavily for the countries that are less immediately classifiable? Do the results reflect a real and stable condition of the countries, or do they depend on decisions that could have been made differently?

1.1 Statements

Given two profiles p and q, we say that p *dominates* q ($p \succeq q$) if p is at least as deprived as q on every single dimension; this relation is reflexive, antisymmetric and transitive, but only partial, since two profiles that are each better on a different dimension simply cannot be compared.

Together with this dominance relation, the distinct profiles observed among the 24 countries form the *poset*, whose *height* (the longest chain of mutually comparable profiles) and *width* (the largest set of mutually incomparable profiles) give a first sense of how much ambiguity the five dimensions generate.

To measure how many countries fall below an acceptable level of deprivation, a *threshold profile* τ is added to the poset: τ represents an "average" country, sitting at the medium grade (grade 1) on all five dimensions at once, and enters as an extra node with no weight of its own, so that every country's profile can be compared against it using the same dominance relation used elsewhere in the analysis.

A *linear extension* of the poset is a complete ranking compatible with all observed dominance relations. For a country c with profile p_c, the *identification degree* is the share of linear extensions in which p_c ranks worse than τ,

$$idn(c) = \Pr_{\ell \sim U(LE(P \cup \{\tau\}))} \Big[p_c \text{ ranks above } \tau \text{ in } \ell \Big],$$

where a value of 1 means the country is deprived in every admissible ranking, 0 means never, and intermediate values reflect genuine ambiguity.

Severity measures the average normalised rank gap above τ, capturing how far beyond the threshold a country typically sits.

Counting linear extensions exactly quickly becomes computationally infeasible, so every quantity above is instead estimated by uniformly sampling a large number of them.

The Data

2.1 Dataset description

This study uses data from the OECD SDG database, covering Goals 1 [1], 4 [2] and 8 [3] and collected through the SDMX API. Where OECD data are permanently missing, a Eurostat fallback is used for four of the five indicators - NEET [5], unemployment [6], low completion [4] and poverty [7] - and two indicators come from other specialised sources, math skills from PISA 2022 [8] and bank account access from the Global Findex 2021 [9]. The analysis starts from seven raw indicators for 24 European OECD countries, each chosen to capture a distinct channel through which young people can be excluded, before two of them are set aside during cleaning.

Monetary **poverty** (ages 18-25) is included because material deprivation is the most direct and least mediated measure of a young person's economic exclusion.

The **NEET** rate captures a broader and more institutionally established condition - disengagement from both education and the labour market at once- and anchors the EU's own Youth Guarantee framework. Youth **unemployment** is kept alongside NEET, despite the definitional overlap between the two (every unemployed young person outside education is by definition also NEET), because it isolates a specific behavioural state, active job search, that a composite NEET rate does not distinguish on its own; whether this overlap is empirically excessive, rather than merely assumed, is decided later directly from the data.

Low math skills and **low reading skills** (PISA) are both included as forward-looking measures of the skill base young people carry into the labour market, rather than a snapshot of their current status; the two are kept as separate indicators at this stage precisely so that their possible redundancy can be tested empirically instead of assumed away.

Non-completion of upper-secondary education* is included because it certifies the minimum credential threshold most commonly used to define labour-market vulnerability.

Lack of a **bank account** (Global Findex) is included as a proxy for financial and digital exclusion, a dimension conceptually distinct from labour-market and educational deprivation.

All seven indicators are oriented so that higher values always mean more deprivation (positively coded variables are inverted using $100 - x$).

From the original seven indicators, only five are kept in the final analysis. Bank account access is dropped not on correlation grounds but because it is nearly saturated across Europe (the large majority of countries sit above the near-universal-access threshold used as a pre-registered cutoff) and therefore fails to discriminate between countries. Low reading skills is dropped after inspecting the full pairwise Spearman correlation matrix of the surviving indicators, because it correlates with low math skills at $\rho = 0.67$, above the 0.6 threshold used elsewhere to flag redundant dimensions; unlike the bank-account removal, this choice was made only after looking at the data, a less rigorous standard than deciding the rule in advance, and is declared as such rather than presented as pre-registered.

The five indicators retained in the final analysis are poverty, NEET, unemployment, low math skills, and low completion of upper-secondary school.

*The EU "early leavers from education and training" (ELET) indicator - ages 18-24, at most lower-secondary education, not in further training - is unavailable in the SDG dataflows used here. As a proxy, this study uses non-completion of upper-secondary education: a related but not identical population, since it is neither age-restricted nor conditional on being out of training.

2.2 Data evaluation

The data preparation process is documented transparently at every stage: rather than simply using the available data, each decision is explained, where every indicator comes from, why certain variables were excluded, and how every cut-off point was determined; each cut-off is also labelled by type (official EU target, empirical average, or normative choice), making it possible to see clearly how subjective each threshold is, rather than presenting arbitrary decisions as if they were objective.

Recoding all indicators so that higher values always indicate greater deprivation further reduces the risk of sign errors in the subsequent poset construction, and makes the indicator matrix easier to interpret consistently. This transparency, however, does not remove every weakness from the underlying data.

The main issue is that the indicators are not perfectly aligned in time, and are not always sourced the same way across countries: labour market data comes from 2023-24, PISA data from 2022, and poverty data from 2021-23, so the resulting matrix is a snapshot built from slightly different reference years - an accepted compromise rather than a fully resolved issue.

Missing values are handled through a strict, three-step search before a country is ever dropped: first the most recent value within the primary 2019–2024 OECD window, then, if still missing, the most recent value in a wider 2015-2024 OECD window, and only as a last resort a verified Eurostat equivalent, available for four of the five indicators - NEET (`edat_lfse_20`)[5], unemployment (`une_rt_a`)[6], low completion (`edat_lfse_14`)[4], and poverty (`ilc_li02`)[7]; no value is ever estimated or interpolated.

On the final run only four cells needed this fallback, all on the completion indicator (Czechia, Ireland, Netherlands, Slovakia), and even this last resort is not fully clean: the Eurostat series measures early school leavers aged 18-24 with at most lower-secondary education and not in further training, a related but not identical population to the OECD non-completion measure used for the other 20 countries in that column, which is not restricted to the 18-24 age band nor conditional on being out of training.

Finally, the sample of only 24 countries, limited by the available data for European countries, means that results, particularly correlations and robustness checks, can be sensitive to a small number of extreme cases; jackknife tests were used to address this concern, but conclusions should still be generalised with caution.

Overall, the data preparation is careful and well documented, which is essential given the study's goal of comparing countries without concealing subjective choices behind a single composite score, and the identified weaknesses do not stem from careless data handling but from trade-offs that could not be fully avoided - the use of a proxy indicator, a fallback source for a handful of cells, and misalignment across reference years - stated explicitly rather than left unaddressed.

Methodology

The study moves from the five cleaned indicators to a final classification of European countries. Each indicator is first converted into an ordinal deprivation grade on a 0-2 scale (0 = low deprivation, 1 = medium, 2 = high), using thresholds declared as one of three types - an official EU target, an empirical anchor, or, only where neither is available, a normative convention guided by the literature.

Indicator	Low cut	High cut	Type (low)	Type (high)
NEET	9	12	EU target	empirical
Low math skills (PISA)	15	30	EU target	normative
Youth poverty (AROP)	10	20	normative	empirical
Youth unemployment	10	20	normative	empirical
Low completion	9	15	normative	normative

Table 1: *Cut-points and their anchoring type. A value above the high cut is graded 2 (high deprivation), between the two cuts grade 1 (medium), below the low cut grade 0 (low).*

For NEET, the low cut (9) is the target set for 2030 by the European Pillar of Social Rights Action Plan [10], and the high cut (12) is the pre-pandemic 2019 EU baseline (12.6%), the Action Plan's critical starting level; the indicator covers ages 15–24 rather than the Action Plan's 15–29, a mismatch justified by the Council Recommendation on the reinforced Youth Guarantee [11], which confirms 15–24 as the historical core of EU youth policy.

For low math skills, the low cut (15) is the 2030 target of the Council Resolution on the European Education Area strategic framework [12], and the high cut (30) sits just above the post-PISA-2022 EU average of ∼29.5% [13] - twice the target rather than a further empirical anchor, since by 2022 the EU average itself had already drifted close to where the high cut was set.

Youth poverty has no official EU percentage target, since the EPSR poverty target is an absolute headcount (−15 million AROPE persons) [10], so both cuts are anchored differently: 20 is the EU average at-risk-of-poverty rate for young people aged 15–29 (∼19.3%) [7], and 10 is about half that average, isolating the best performers. The OECD series used here follows the 50%-of-median poverty convention while the Eurostat AROP statistic is 60%-of-median, so this empirical anchor is approximate and conservative.

Youth unemployment has no isolated EPSR target either: 20 is clearly above the EU 15–24 average (∼14.5% in 2023) [6], and 10 is below that average, isolating structurally stable youth labour markets.

Low completion uses two normative conventions: the lower cut (9) borrows the EU early-leavers target ($< 9\%$), set by the European Education Area strategic framework [12], as the nearest institutional anchor, even though the measured concept (100 minus the upper-secondary completion rate) is systematically larger than the EU early-leaver share [4] - a deliberate, declared choice that makes these cuts stricter than the target they borrow from, and pushes several otherwise well-performing countries into grade 1–2 on this dimension.

Instead of averaging these grades into a single score, country profiles are compared through the dominance relation introduced above and organised into a poset: a country is more deprived than another only if it is at least as deprived on every dimension, and countries that are better on different dimensions are left incomparable rather than forced into a ranking.

For each country, an identification degree (how consistently it ranks worse than τ) and a severity score (how far beyond τ) are computed, both defined above; the poset's structural properties (height, width, share of incomparable pairs) are reported separately, since they measure ambiguity independently of any threshold.

Because counting linear extensions exactly is infeasible (see *Tools*, below), every one of these quantities is instead estimated by uniformly sampling a large number of them (20000 samples); to guard against implementation errors, each result was computed twice - once with `parsec`, and once with an independent implementation written from scratch for this project - both first validated on a small toy poset whose identification degrees could be worked out by hand, and only afterwards applied to the real data, with the two sets of results compared through their Spearman correlation to confirm they agreed.

The resulting poset ranking is then compared with a conventional composite index, built as a benchmark by normalising each of the five indicators to the $[0, 1]$ range and taking a simple average; this composite ranking is compared to the poset's average rank, each country's mean position across the sampled linear extensions, using Spearman correlation, with particular attention to the countries whose position shifts most between the two approaches.

Finally, five checks assess which results are stable and which depend on specific methodological choices: an indicator jackknife repeats the whole analysis five times, leaving out one dimension at a time; a country jackknife does the same 24 times, leaving out one country at a time; three alternative discretisation schemes are tested, the tercile-based Schema B (sensitivity to the normative cut-points of Schema A) and a finer, quintile-based Schema C (sensitivity to the number of grade levels used); and a permutation test shuffles each indicator's values independently across countries (1,000 repetitions) to see whether the incomparability observed in the real data reflects a genuine relationship between the dimensions or could just as easily be due to chance.

A last, complementary analysis identifies, for every country with an identification degree below 1, which dimension (or dimensions) kept its grade below τ - in other words, what stood in the way of it being certainly deprived; counting how often each dimension plays this "blocking" role across countries shows which single dimension is most often responsible for keeping a country's classification uncertain.

Results and Comments

Correlation among the indicators

The five retained indicators show a median absolute pairwise Spearman correlation of 0.27 across the 24 countries. The strongest association is between *poverty* and *low_completion* ($\rho = 0.57$), followed by *neet* and *unemp* (0.42) and *unemp* and *lowskill_math* (0.41); no pair reaches the $\rho > 0.6$ redundancy threshold used in the cleaning step, although poverty and low_completion sit close to it, and the overlap between neet and unemp, flagged as a potential problem when the indicator set was defined, turns out to be well within the bar the cleaning rule had set in advance.

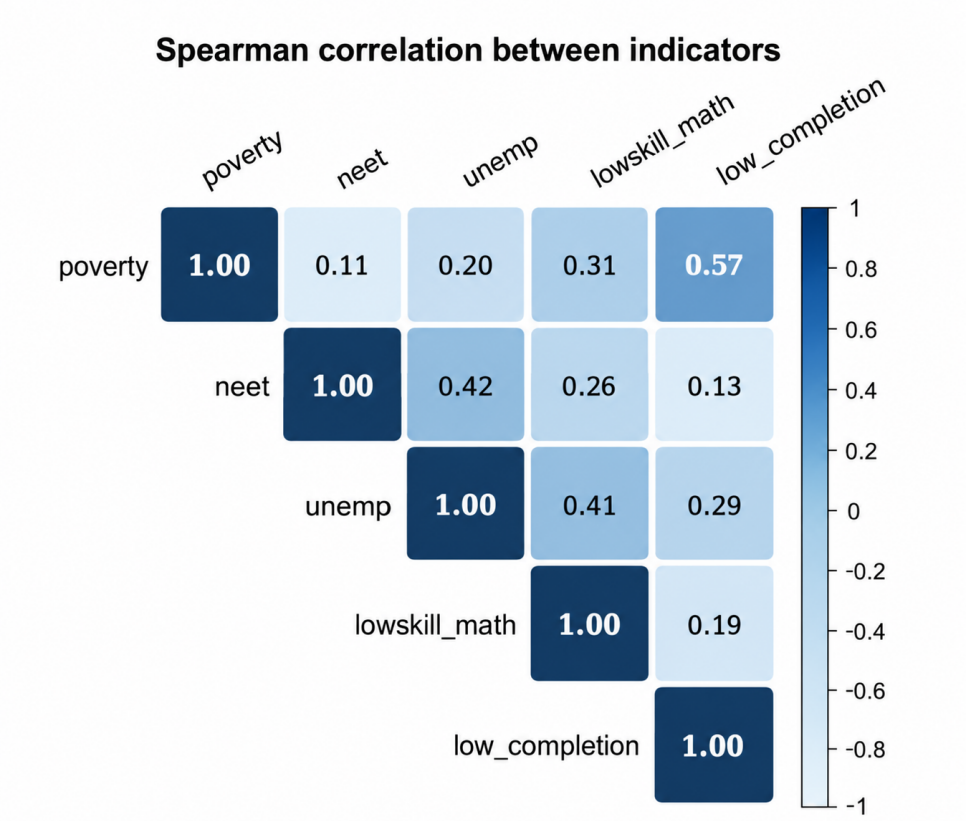


Figure 1: *Spearman correlation matrix among the five retained indicators.*

This result is the empirical basis for the choice of method, not merely a preliminary check: with correlations this moderate, the five dimensions carry substantially independent information, so a country can be badly off on one and comparatively fine on another. A composite index, being an average, would let good performance on one dimension compensate for a severe weakness on another - precisely the property that makes it a poor fit here.

The poset avoids this by construction, ranking a country as more deprived only when it is at least as deprived on every dimension simultaneously, and leaving genuinely divergent profiles incomparable rather than forcing them into a single score.

Had the indicators instead been strongly correlated, this advantage would have largely disappeared, not because the poset would stop working, but because it would stop adding anything: correlated dimensions move together, so almost every pair of countries would already be comparable by dominance alone, and a simple average would recover essentially the same ordering the poset does - at that point the extra computational machinery would no longer be justified by the data.

Poset structure and the Hasse diagram

The 24 country profiles collapse to 18 distinct profiles, with four exact ties: Austria, Lithuania and the United Kingdom share one profile; Belgium and Ireland share another; the Czech Republic, Slovenia and Switzerland share a third; France and Poland share a fourth. The threshold profile $\tau = (1, 1, 1, 1, 1)$ is not observed among them and enters the poset as a nineteenth node carrying zero weight. Of the 153 pairs of distinct country profiles, 56.2% are mutually incomparable; over the 276 country pairs the share falls to 44.2%, because the exact ties concentrate among the comparable profiles at the bottom of the order. More than half of the comparisons the data allow between distinct profiles therefore cannot be resolved by dominance alone, which is precisely the information a scalar index would discard.

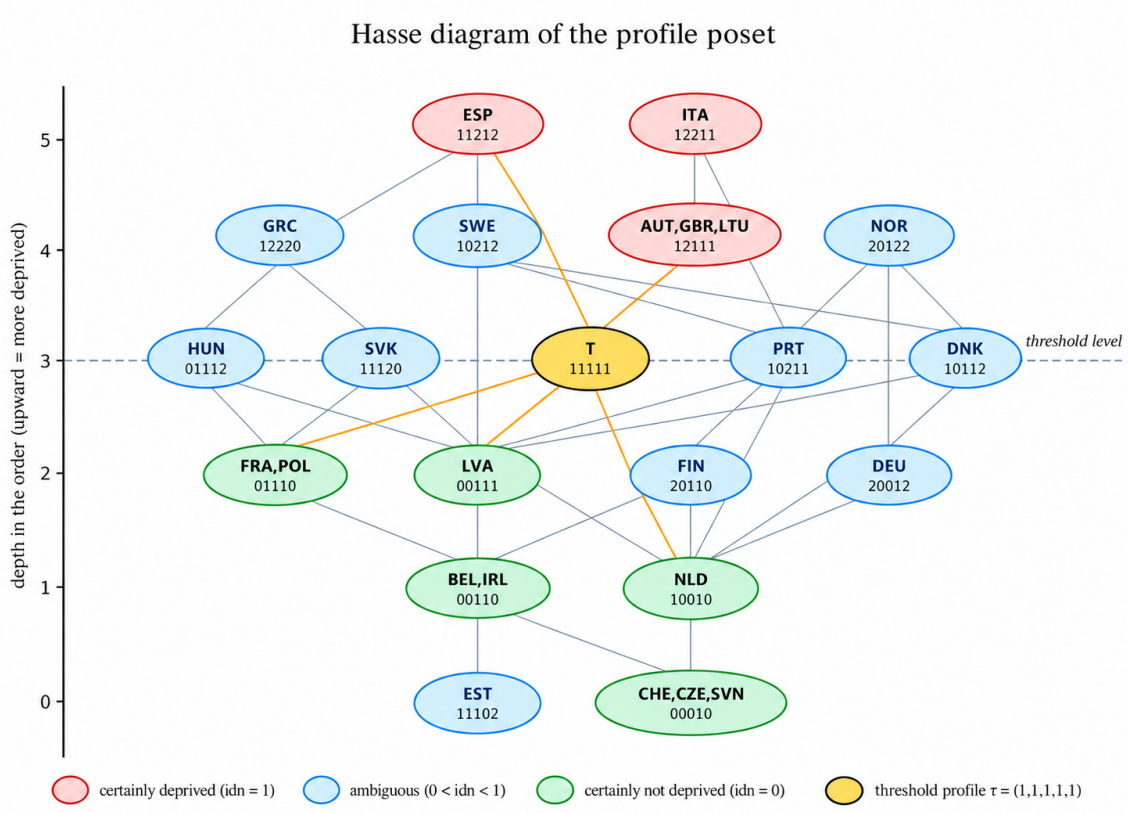


Figure 2: *Hasse diagram of the profile poset under Schema A. Nodes are the 18 distinct country profiles, labelled with the countries sharing them; edges are cover relations; vertical position is depth in the order (upward = more deprived). The threshold profile τ is shown in orange.*

Figure 2 draws the resulting order. Nodes are distinct profiles, labelled with the countries sharing them and with their grade vector below; edges are cover relations, so relations implied by transitivity are not drawn twice; vertical position is depth in the order. The diagram is deterministic, built from component-wise dominance alone, and it is the structure on which the linear extensions of the next section are sampled. It makes the three-way partition immediate: three profiles lie strictly above τ, accounting for the five certainly deprived countries; five lie strictly below it, covering the nine countries at idn = 0; the remaining ten countries occupy profiles incomparable with τ, worse than the medium grade on some dimension and better on another.

Two features deserve comment. The widest level of the order is the one containing τ: ambiguity is not spread evenly, it concentrates exactly where the threshold sits, the structural counterpart of the 44.2% incomparability figure. And depth is not severity - the order has four maximal profiles (Spain, Italy, Greece, Norway) rather than one, so no single worst performer is identified, while Estonia sits at the very bottom, its unique grade 0 in mathematics leaving nothing below it, yet its identification degree is 0.474 against the 0 of the Czech Republic, Slovenia and Switzerland at the same depth.

Identification degree and severity

As we can see from Table 2, five countries are certainly deprived (idn = 1): Spain, Italy, Austria, Lithuania and the United Kingdom. Nine are certainly not deprived (idn = 0): Belgium, the Czech Republic, France, Ireland, Latvia, the Netherlands, Poland, Slovenia and Switzerland. The remaining ten fall strictly between the two extremes, ranging from Estonia (0.474) to Norway (0.947). The intermediate group is not spread evenly across that interval: six of the ten sit between 0.47 and 0.55, close to the point of maximum ambiguity, while Sweden, Greece and Norway cluster above 0.88, near-certain without reaching certainty. The spread within the group is worth noting in its own right: idn ranges across nearly the entire interval, which means "ambiguous" is not a single condition but a continuum running from countries barely more likely than not to be deprived to countries deprived under almost every admissible ranking. Norway is the extreme case, kept below 1 by a single grade-0 dimension out of five.

Severity, which measures how far beyond τ a country's profile typically sits rather than merely whether it lies above it, adds information that idn alone does not capture. It is zero for the nine countries at idn = 0, since they never rank above τ to begin with. Among the ten intermediate countries it rises broadly in step with idn, from Denmark and Portugal (≈ 0.14) up to Norway (0.709), though not monotonically: Estonia has the lowest identification degree of the group (0.474) and yet a higher severity (0.214) than either, because on the minority of rankings where it does exceed τ it exceeds it by a wide margin.

The two measures come apart most clearly at the top. Among the five certainly-deprived countries idn stops discriminating – all five sit at exactly 1 – while severity still does: Austria, Lithuania and the United Kingdom clear the threshold by a comparatively modest margin (0.385), whereas Spain (0.858) and Italy (0.771) are deprived far more severely on top of being deprived with certainty. Norway makes the point from the other side: at idn = 0.947 it is not certainly deprived, yet its severity (0.709) is nearly twice that of the Austria–Lithuania–United Kingdom profile. Certainty of identification and depth of deprivation are distinct quantities, and reporting only the first would collapse them. The ordering of severity among the certainly-deprived countries also has a practical reading: if resources had to be prioritised within this group of five, Spain and Italy would be the first call on that basis alone, since their profiles are not merely certain but severe, while Austria, Lithuania and the United Kingdom clear the bar without being deeply beyond it.

Table 2 reports the full country-by-country results, ordered from the least to the most deprived.

Cross-validation of the two implementations

Both columns of identification degrees in Table 2 are Monte Carlo estimates: the from-scratch implementation samples 20,000 linear extensions through a Bubley–Dyer chain, while `parsec` runs its own sampler at a tolerance of 10^{-3}. Neither is an exact enumeration, so agreement between them is evidence about the code, not about the mathematics of the method.

The two sets of estimates never differ by more than 0.055 (Estonia, 0.474 against 0.529), and their Spearman correlation is 0.9861. The rank correlation is reported alongside the maximum absolute difference because the two measure different kinds of agreement: the former checks that the two implementations order the countries the same way, the latter that they agree on the numbers themselves, and a method could in principle pass one check while failing the other.

Country	Profile	idn (`parsec`)	idn (from scratch)	Severity
BEL	0,0,1,1,0	0.000	0.000	0.000
CHE	0,0,0,1,0	0.000	0.000	0.000
CZE	0,0,0,1,0	0.000	0.000	0.000
FRA	0,1,1,1,0	0.000	0.000	0.000
IRL	0,0,1,1,0	0.000	0.000	0.000
LVA	0,0,1,1,1	0.000	0.000	0.000
NLD	1,0,0,1,0	0.000	0.000	0.000
POL	0,1,1,1,0	0.000	0.000	0.000
SVN	0,0,0,1,0	0.000	0.000	0.000
EST	1,1,1,0,2	0.474	0.529	0.214
DNK	1,0,1,1,2	0.486	0.491	0.139
PRT	1,0,2,1,1	0.498	0.473	0.141
DEU	2,0,0,1,2	0.525	0.480	0.213
FIN	2,0,1,1,0	0.531	0.531	0.213
SVK	1,1,1,2,0	0.554	0.527	0.202
HUN	0,1,1,1,2	0.689	0.671	0.308
GRC	1,2,2,2,0	0.885	0.856	0.589
SWE	1,0,2,1,2	0.886	0.895	0.497
NOR	2,0,1,2,2	0.947	0.942	0.709
AUT	1,2,1,1,1	1.000	1.000	0.385
ESP	1,1,2,1,2	1.000	1.000	0.858
GBR	1,2,1,1,1	1.000	1.000	0.385
ITA	1,2,2,1,1	1.000	1.000	0.771
LTU	1,2,1,1,1	1.000	1.000	0.385

Table 2: Identification degree and severity by country. *The identification degree is the share of sampled linear extensions in which a country's profile ranks above the threshold $\tau = (1, 1, 1, 1, 1)$, estimated independently with* `parsec` *and with the from-scratch implementation described in Section 3; severity measures how far beyond τ a country's profile typically sits. Countries are ordered from the least to the most deprived. Profiles are written as (poverty, neet, unemp, lowskill_math, low_completion), each on a 0–2 grade scale.*

Comparison with the composite index

The poset average rank and the equal-weight composite index agree closely on the overall ordering ($\rho = 0.85$). The disagreements, however, are concentrated and systematic rather than scattered: nine countries shift by at least three positions (Figure 3), splitting cleanly into two groups by the shape of their profile.

The first group is penalised by the poset and spared by the composite index: countries with one severe weakness among otherwise moderate scores. Austria moves nine positions, from 9th on the composite index to 18th under the poset, and the United Kingdom eight; both carry the profile $(1, 2, 1, 1, 1)$, where a grade-2 NEET rate is averaged away against four moderate scores. Slovakia $(1, 1, 1, 2, 0)$ moves four positions for the same reason. The second group is the mirror image: deprivation spread across several dimensions without ever becoming acute. Denmark $(1, 0, 1, 1, 2)$ moves nine positions the other way – the composite index ranks it 20th, marking down its high non-completion figure, while the poset keeps it ambiguous (idn $= 0.486$) because a grade-0 NEET rate blocks dominance over τ. France $(0, 1, 1, 1, 0)$ is the clearest case: mediocre on three dimensions and good on two, it sits 13th on the composite index yet is dominated by τ outright, so the poset places it among the certainly non-deprived. Estonia and Portugal shift four positions each on the same logic.

Greece is worth isolating: it is the one country the composite index ranks as the single most deprived in the sample (24th) while the poset stops short of certainty (idn $= 0.885$). Its profile $(1, 2, 2, 2, 0)$ carries three grade-2 scores, more than any certainly-deprived country, but its grade-0 on completion is enough to break dominance – the two methods apply different decision rules to the same profile, and only the non-compensatory one treats a single good dimension as sufficient to withhold a verdict.

One divergence, finally, is not attributable to aggregation at all. Austria, the United Kingdom and Lithuania share an identical grade profile and therefore an identical poset position, yet the composite index separates them by ten positions (9th, 10th and 19th). The gap is a product of discretisation: the composite index works on the continuous values, while the poset sees only the three ordinal bands, and within-band variation is information the ordinal approach deliberately discards.

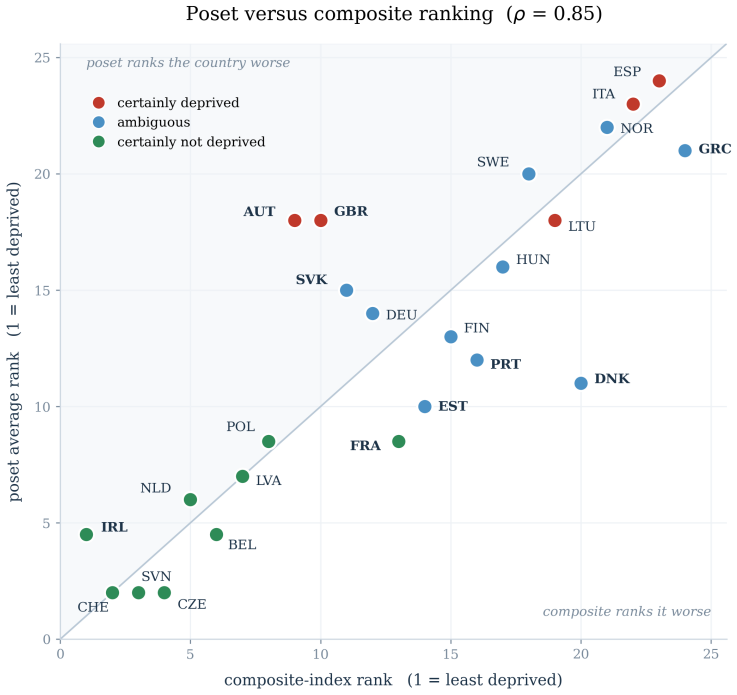


Figure 3: *Composite-index rank versus poset average rank (Spearman $\rho = 0.85$). Countries with a rank shift of at least 3 positions are labelled in red.*

Robustness checks

- **Indicator jackknife.** Dropping one indicator at a time never costs a country its certainty: a profile that beats τ on five dimensions still beats it on any four, and the same holds in reverse. The folds act only on the ambiguous ten, resolving a country whenever the omitted indicator was its *only* sub-threshold dimension, which makes it certainly deprived, or its *only* above-threshold one, which makes it certainly not. Nine of the ten are resolved in at least one fold and only Germany never is; *low_completion* resolves five countries, *neet* four, *poverty* and *lowskill_math* two each, and *unemp* one. Being the sole blocker is much rarer than blocking at all - NEET sits below τ in 13 countries but is the only such dimension in four of them - so the intermediate group is not unstable in the ordinary sense: it is undecidable at five dimensions and decidable at four, and the ambiguity reported here reflects how many dimensions the analysis admits rather than noise in the data.
- **Country jackknife.** Removing any one of the 24 countries moves the others' identification degrees by a few hundredths at most. The certain groups are untouched, but this holds by construction rather than as a finding: dominance with respect to τ depends only on a country's own profile, not on who else is in the sample. The check confirms that the implementation behaves as the theory requires.
- **Number of grade levels.** The share of incomparable country pairs rises from 44% under Schema A to 53% under Schema B (both 3 levels) and 67% under the quintile-based Schema C (Figure 4). Part of the observed incomparability is therefore a consequence of how finely the ordinal scale is cut rather than a pure property of the data: a finer scale records more distinctions, and more distinctions mean more pairs differing in opposite directions.

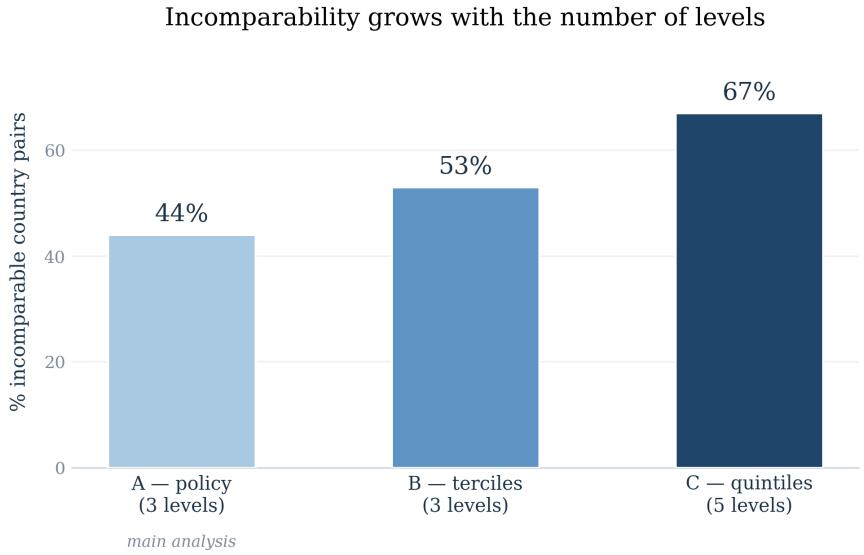


Figure 4: *Share of incomparable country pairs under Schema A (policy-anchored, 3 levels), Schema B (terciles, 3 levels) and Schema C (quintiles, 5 levels).*

- **Permutation test.** Shuffling each indicator independently across countries (1,000 replications) preserves each marginal distribution while destroying the association between dimensions. The null distribution of the incomparable-pair share is centred at $63.5\% \pm 7.1\%$ against an observed 44.2%, which lies below almost the entire null ($p \approx 0.003$; equivalently $p = 0.997$ for the test in the opposite direction). The real data are markedly *more* ordered than independent indicators would be (Figure 5).

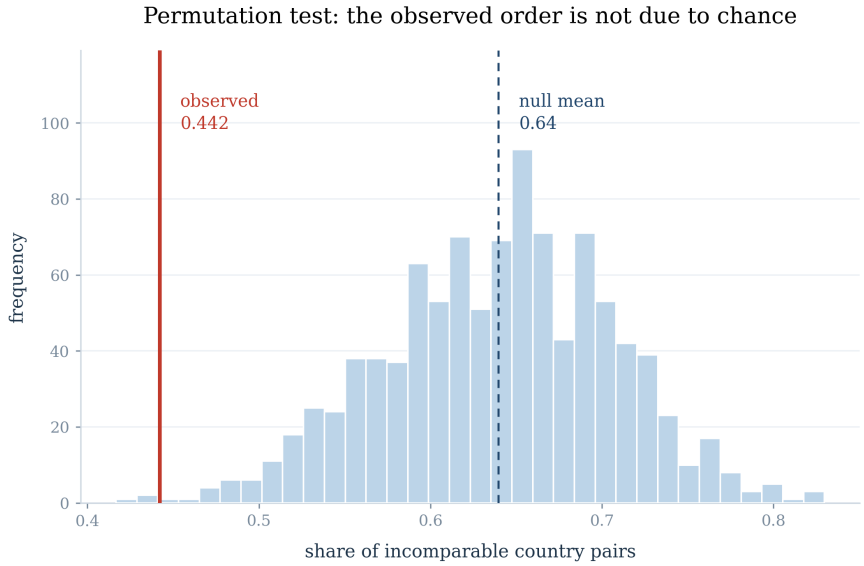


Figure 5: *Null distribution of the incomparable-pair share under 1,000 column-wise permutations of the five indicators, against the observed value (red line). The observed* 44.2% *lies below almost the entire null distribution, which is centred at* 63.5%*.*

Which dimensions block certain identification

A country fails to be certainly deprived as soon as one of its grades falls below the corresponding component of τ, since a single sub-threshold dimension breaks dominance. Among the 19 countries with idn < 1, *neet* is the most frequent blocking dimension (13 countries), followed by *low_completion* (11), *poverty* (9), *unemp* (5) and *lowskill_math* (1) (Table 3).

The ordering follows from the grade distributions rather than from any property of the poset, and it must not be read as a ranking of policy urgency. NEET blocks most often precisely because it is the dimension on which most countries already perform well: a good NEET rate is simply the most widely available way of escaping an otherwise certain classification. *lowskill_math* blocks only Estonia for the opposite reason - almost no country meets the EU target, so the dimension rescues nobody. Norway, Greece and Sweden illustrate the mechanism at its sharpest, each held below certainty by exactly one grade-0 dimension out of five.

Dimension	Countries below threshold
neet	13
low_completion	11
poverty	9
unemp	5
lowskill_math	1

Table 3: *Number of countries (out of the 19 with* idn < 1*) below the threshold on each dimension.*

Comments

The results above establish which countries are certainly deprived and which are not; they say nothing yet about whether the countries usually grouped together - by geography, shared history, or a common institutional model - end up close together in the dominance order as well. Four such groupings are checked below against the actual profiles.

Take first the countries routinely described as sharing a single Mediterranean youth crisis: Spain, Italy, Greece and Portugal. On one dimension this holds exactly - all four score grade 2 on unemployment, the only point where the whole group converges. Elsewhere it does not. Portugal, with a NEET score of 0 that matches the Nordic countries rather than its regional neighbours, stays outside the clearly deprived cluster (idn = 0.498), while Spain and Italy fall clearly within it. Yet even these last two, despite sharing the same outcome (idn = 1), turn out to be mutually incomparable: their profiles - Spain $(1, 1, 2, 1, 2)$ and Italy $(1, 2, 2, 1, 1)$ - diverge, since Spain performs worse on school completion and Italy on NEET. The two countries arrive at the same threshold by two different routes, not one shared crisis.

Much the same happens where the shared trait is history rather than geography. Estonia, Latvia and Lithuania went through withdrawal from the USSR, market reforms in the 1990s and EU accession in 2004 together, which makes them the most direct check available of whether common history predicts a common outcome. It does not. Latvia is certainly not deprived (idn = 0), Lithuania is certainly deprived (idn = 1), and Estonia is genuinely ambiguous (idn = 0.474) - the only country in the sample to meet the EU target on mathematics while simultaneously failing it on completion. The comparative structure is sharper still: Estonia $(1, 1, 1, 0, 2)$ is incomparable with both of its neighbours, better on some dimensions and worse on others than each, while Latvia and Lithuania are comparable to one another, the Lithuanian profile $(1, 2, 1, 1, 1)$ dominating the Latvian one $(0, 0, 1, 1, 1)$ on every single component. Three decades of shared history produced one ordering relation out of three possible pairs.

Nor does a shared welfare model fare any better. Norway, Denmark, Sweden and Finland do converge on one dimension - all four sit at grade 0 on NEET, and no other group in the sample keeps more young people in education or work - so a generous welfare state, on this evidence, delivers engagement reliably. It does not deliver the rest as reliably: none of the four reaches grade 0 on poverty, and three of the four reach grade 2 on completion, as if the Nordic model traded formal certification for immediate engagement. Nor does the group form a simple ranking internally. Norway $(2, 0, 1, 2, 2)$ dominates both Denmark $(1, 0, 1, 1, 2)$ and Finland $(2, 0, 1, 1, 0)$, worse or equal on every dimension than each, but is incomparable with Sweden $(1, 0, 2, 1, 2)$: Sweden is better on poverty and worse on unemployment. Denmark and Finland are incomparable with one another, and so are Sweden and Finland.

Finally, even a shared institutional design does not travel well. Austria and Germany are the two countries most often cited as reference cases for dual vocational training, a system meant to protect specifically against NEET risk. Germany confirms the expectation, at grade 0 on NEET; Austria contradicts it as sharply as the data allow, with a single grade 2 on NEET sufficient on its own to make it certainly deprived despite an otherwise mid-range profile. The two profiles, Austria $(1, 2, 1, 1, 1)$ and Germany $(2, 0, 0, 1, 2)$, are incomparable: Austria is worse on NEET and unemployment, Germany worse on poverty and completion. This is not a difference of degree between two countries built on the same model - it is two opposite patterns of disadvantage produced by that model.

None of the four groupings survives intact. Within almost every one of them - Spain and Italy, Estonia and its Baltic neighbours, Austria and Germany - there is at least one pair of countries that turns out to be mutually incomparable, a sign that narrative proximity does not imply any ordering relation in the data. This, more than any single case, is the strongest argument in favour of the poset that the country-level evidence offers: the similarities that matter here are those of profile, not of region, history or welfare model, and they have to be found there, not assumed.

Conclusion

Answers to the research questions

The three sub-questions set out in the introduction are answered in turn below, before returning to the general question.

1. **Which countries are unambiguously "left behind", and which are not?** Both groups exist, and they are smaller than the sample. Five countries - Spain, Italy, Austria, Lithuania and the United Kingdom - are certainly deprived: their profiles dominate the reference threshold on every dimension at once, so they rank above it in every admissible ranking. Nine countries - Belgium, Switzerland, the Czech Republic, France, Ireland, Latvia, the Netherlands, Poland and Slovenia - are certainly not: the threshold dominates them, so they never rank above it. The answer is therefore affirmative in both directions, but only for 14 of the 24 countries. The remaining ten are incomparable with the threshold - worse on some dimension, better on another - and for them no arrangement of the five dimensions settles the matter. Severity adds a second layer to the certain group that identification alone would flatten: Spain (0.858) and Italy (0.771) are deprived far more deeply than Austria, Lithuania and the United Kingdom (0.385), even though all five are deprived with equal certainty.

2. **Which countries are classified differently by the two methods, and why?** Nine countries shift by at least three positions, and the shifts are systematic rather than random. Countries with one severe weakness among moderate scores are penalised by the poset and spared by the composite index: Austria moves nine positions and the United Kingdom eight, both carrying a grade-2 NEET rate that averaging dilutes against four moderate scores. Countries whose disadvantage is spread thinly without ever becoming acute move the opposite way: Denmark shifts nine positions, marked down by the composite index for a single poor indicator while the poset keeps it ambiguous, and France is ranked mid-table by the composite index yet dominated by the threshold outright. The overall correlation ($\rho = 0.85$) shows the two methods agree for most of the sample; the exceptions are not noise but the direct consequence of one explicit assumption - whether a good result on one dimension may compensate a bad one on another.

3. **How robust are these results, and which dimension weighs most?** The two answers point in opposite directions. The certain core is stable: it survives the removal of any single country or any single indicator, since dominance with respect to τ cannot be lost by dropping a dimension, only gained. The intermediate group is not stable in that sense, but it is not unstable either - it is undecidable at five dimensions and decidable at four, resolved fold by fold whenever the omitted indicator was a country's only sub-threshold dimension; nine of the ten ambiguous countries are resolved in at least one fold, only Germany in none. The switch from policy-anchored to tercile-based cut-points reshuffles the group substantially ($\rho \approx 0.60$), and incomparability itself rises from 44% to 67% as the ordinal scale is refined from three levels to five. The fragility, in every form it takes, falls entirely on the cases that were already least classifiable, never on the extremes. As for the decisive dimension, the bottleneck analysis identifies NEET, which keeps 13 of the 19 non-certain countries below the threshold - not because it is the most critical dimension in Europe, but for the opposite reason: it is the one on which most countries already perform well, and therefore the one most often available to spare a country from an otherwise certain classification.

In response to the general question, a unique classification of European countries is possible only for a limited, well-defined subset of the sample, while for the majority the answer depends substantially on the aggregation method adopted. This is not a shortcoming of the study but its central finding: the choice between a composite index and a non-compensatory method such as the poset carries a concrete impact on which countries a policymaker would treat as priorities. The poset does not resolve the ambiguity affecting the intermediate group; it refuses to conceal it, which an averaged score would have done automatically and invisibly.

Strengths/weaknesses of the study

- The main strength of the study is its methodological validation: two independent implementations of the poset method (from scratch and `parsec`), verified first on small posets with hand-computable or analytically known solutions and then cross-checked against each other on the real data, ensure that the results are not an artifact of coding errors.
- A second strength is the depth of the robustness checks: five distinct tests, together with the bottleneck analysis, make it possible to state explicitly which results are stable and which depend on specific methodological choices, rather than presenting the whole framework as equally reliable.
- A third strength is the transparency of the threshold system itself. Every cut-off used to build the ordinal scale is labelled by one of three declared types - an official EU policy target, an empirical anchor calibrated on EU averages, or, only where neither is available, an explicit normative convention - so the extent to which any given cut-off rests on a policy benchmark rather than on judgement is always traceable. This is a genuine advantage over composite indices, where comparable choices are typically embedded in undisclosed weights rather than stated openly.
- The most significant limitation concerns the sensitivity of the intermediate group to the choice of indicators: nine of the ten ambiguous countries are resolved into a certain classification by at least one indicator jackknife fold, which means their status in the main analysis reflects the particular set of five dimensions retained as much as it reflects their actual condition - precisely where the poset approach would otherwise have the greatest added value over a composite index.
- A second limitation concerns the normative nature of the thresholds themselves. Traceable labelling notwithstanding, both the reference threshold τ and several of the Schema A cut-offs - wherever no official EU target or reliable empirical average was available - rest on an explicit normative convention rather than an external benchmark, and remain judgement calls that influence every downstream result.

Directions for further research or improvements

A first natural step would be to extend the analysis to a temporal dimension, moving from a single cross-sectional picture to a panel analysis spanning multiple years, in order to verify whether the pattern of incomparability among European countries observed here is stable over time, or whether, on the contrary, countries currently classified as ambiguous are gradually shifting toward a state of definite deprivation or definite non-deprivation. A second direction would expand the number of dimensions considered, including aspects of youth well-being not covered in this study, such as mental health, access to housing, or forms of social and digital exclusion, to test whether the observed pattern of deprivation holds once the analysis moves beyond the five economic and educational dimensions used here. A third direction would apply the same methodological approach to a broader geographic context, extending the comparison beyond the 24 European countries considered here - to the entire OECD sample, for instance, or to non-European countries with comparable welfare systems - to understand whether European youth deprivation is a specific case or reflects broader trends.

References

[1] OECD. SDG data explorer — goal 1: Poverty indicators. Dataflow `OECD.WISE.RSB / DSD_SDG@DF_SDG_G_1`, data-explorer.oecd.org, 2026. Accessed September 2026.

[2] OECD. SDG data explorer — goal 4: Education indicators. Dataflow `OECD.WISE.RSB / DSD_SDG@DF_SDG_G_4`, data-explorer.oecd.org, 2026. Accessed September 2026.

[3] OECD. SDG data explorer — goal 8: Decent work and economic growth indicators. Dataflow `OECD.WISE.RSB / DSD_SDG@DF_SDG_G_8`, data-explorer.oecd.org, 2026. Accessed September 2026.

[4] Eurostat. Early leavers from education and training by sex and labour status. Eurostat Data Browser, dataset `edat_lfse_14`, 2026. Accessed September 2026.

[5] Eurostat. Young persons neither in employment nor in education and training by labour status (NEET rates). Eurostat Data Browser, dataset `edat_lfse_20`, 2026. EU-LFS. Accessed September 2026.

[6] Eurostat. New lows for EU unemployment in 2024. Eurostat News, based on dataset `une_rt_a`, ec.europa.eu/eurostat/web/products-eurostat-news, 2025. Youth unemployment rate (age 15–24), EU-LFS: 14.9% in 2024, up 0.4 pp from 2023 (i.e. ∼14.5% in 2023). Accessed September 2026.

[7] Eurostat. Young people — social inclusion. Statistics Explained, based on dataset `ilc_li02`, ec.europa.eu/eurostat/statistics-explained, 2026. At-risk-of-poverty rate for people aged 15–29 in the EU: 19.3% in 2022. Accessed September 2026.

[8] OECD. PISA 2022 results (volume I): The state of learning and equity in education. Technical report, OECD Publishing, Paris, 2023.

[9] Asli Demirgüç-Kunt, Leora Klapper, Dorothe Singer, and Saniya Ansar. The global findex database 2021: Financial inclusion, digital payments, and resilience in the age of COVID-19. Technical report, World Bank, Washington, DC, 2022.

[10] European Commission. The european pillar of social rights action plan. Technical Report COM(2021) 102 final, Publications Office of the European Union, Luxembourg, 2021. NEET rate (15–29) to fall from 12.6% in 2019 to 9% by 2030. EUR-Lex CELEX 52021DC0102.

[11] Council of the European Union. Council recommendation of 30 october 2020 on a bridge to jobs — reinforcing the youth guarantee. Technical Report 2020/C 372/01, Official Journal of the European Union, 2020. OJ C 372, 4.11.2020, pp. 1–9. Confirms ages 15–24 as the historical core of EU youth policy. EUR-Lex CELEX 32020H1104(01).

[12] Council of the European Union. Council resolution on a strategic framework for european cooperation in education and training towards the european education area and beyond (2021–2030). Technical Report 2021/C 66/01, Official Journal of the European Union, 2021. OJ C 66, 26.2.2021, pp. 1–21. Sets EU-level 2030 targets including: share of low-achieving 15-year-olds in basic skills below 15%; share of early leavers from education and training below 9%. EUR-Lex CELEX 32021G0226(01).

[13] Eurostat. SDG 4 — quality education. Statistics Explained, ec.europa.eu/eurostat/statistics-explained, 2026. 29.5% of EU 15-year-olds were low achievers in mathematics in PISA 2022, against the EU 2030 target of below 15%. Accessed September 2026.

[14] Marco Fattore. Partially ordered sets and the measurement of multidimensional ordinal deprivation. *Social Indicators Research*, 128:835–858, 2016.

[15] Marco Fattore and Alberto Arcagni. PARSEC: An R package for poset-based evaluation of multidimensional poverty. In Rainer Brüggemann, Lars Carlsen, and Jochen Wittmann, editors, *Multi-indicator Systems and Modelling in Partial Order*, pages 317–330. Springer, New York, 2014.

[16] Marco Fattore and Alberto Arcagni. A reduced posetic approach to the measurement of multidimensional ordinal deprivation. *Social Indicators Research*, 136(3):1053–1070, 2018.

[17] Paola Annoni and Rainer Brüggemann. Exploring partial order of european countries. *Social Indicators Research*, 92:471–487, 2009.

[18] Marco Fattore and Leonardo Salvatore Alaimo. A partial order toolbox for building synthetic indicators of sustainability with ordinal data. *Socio-Economic Planning Sciences*, 89:101623, 2023.

[19] Rainer Brüggemann and Lars Carlsen, editors. *Partial Order in Environmental Sciences and Chemistry*. Springer, Berlin, Heidelberg, 2006.

[20] Russ Bubley and Martin Dyer. Faster random generation of linear extensions. *Discrete Mathematics*, 201(1–3):81–88, 1999.

[21] Graham Brightwell and Peter Winkler. Counting linear extensions. *Order*, 8:225–242, 1991.

[22] Sabina Alkire and James Foster. Counting and multidimensional poverty measurement. *Journal of Public Economics*, 95(7–8):476–487, 2011.

[23] Martin Ravallion. Mashup indices of development. *The World Bank Research Observer*, 27(1):1–32, 2012.

[24] Koen Decancq and María Ana Lugo. Weights in multidimensional indices of wellbeing: An overview. *Econometric Reviews*, 32(1):7–34, 2013.

[25] OECD and JRC European Commission. *Handbook on Constructing Composite Indicators: Methodology and User Guide*. OECD Publishing, Paris, 2008.

[26] OECD. The short and winding road to 2030: Measuring distance to the sdg targets. Technical report, OECD Publishing, Paris, 2022.